



Tingli Dazao Xiefei Decoction ameliorates asthma *in vivo* and *in vitro* from lung to intestine by modifying NO–CO metabolic disorder mediated inflammation, immune imbalance, cellular barrier damage, oxidative stress and intestinal bacterial disorders

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ABSTRACT

Ethnopharmacological relevance: Asthma is a chronic airway inflammatory disease. Current treatment of mainstream medications has significant side effects. There is growing evidence that the refractoriness of asthma is closely related to common changes in the lung and intestine. The lungs and intestines, as sites of frequent gas exchange in the body, are widely populated with gas signaling molecules NO and CO, which constitute NO–CO metabolism and may be relevant to the pathogenesis of asthma in the lung and intestine. The Chinese herbal formula Tingli Dazao Xiefei Decoction (TD) is commonly used in clinical practice to treat asthma with good efficacy, but there are few systematic evaluations of the efficacy of asthma on NO–CO metabolism, and the mode of action of its improving effect on the lung and intestine is unclear.

Aim of the study: To investigate the effect of TD on the lung and intestine of asthmatic rats based on NO–CO metabolism.

Materials and methods: *In vivo*, we established a rat asthma model by intraperitoneal injection of sensitizing solution with OVA atomization, followed by intervention by gavage administration of TD. We simultaneously examined alterations in basal function, pathology, NO–CO metabolism, inflammation and immune cell homeostasis in the lungs and intestines of asthmatic rats, and detected changes in intestinal flora by macrogenome sequencing technology, with a view to multi-angle evaluation of the treatment effects of TD on asthmatic rats. *In vitro*, lung cells BEAS-2B and intestinal cells NCM-460 were used to establish a model of lung injury causing intestinal injury using LPS and co-culture chambers, and lung cells or intestinal cells TD-containing serum was administered to intervene. Changes in inflammatory, NO–CO metabolism-related, cell barrier-related and oxidative stress indicators were measured in lung cells and intestinal cells to evaluate TD on intestinal injury by way of amelioration and in-depth mechanism.

Results: *In vivo*, our results showed significant basal functional impairment in the lung and intestine of asthmatic rats, and an inflammatory response, immune cell imbalance and intestinal flora disturbance elicited by NO–CO metabolic disorders were observed ($P < 0.05$ or 0.01). The administration of TD was shown to deliver a multidimensional amelioration of the impairment induced by NO–CO metabolic disorders ($P < 0.05$ or 0.01). *In vitro*, the results showed that LPS-induced lung cells BEAS-2B injury could cause NO–CO metabolic disorder-induced inflammatory response, cell permeability damage and oxidative stress damage in intestinal cells NCM-460 ($P < 0.01$). The ameliorative effect on intestinal cells NCM-460 could only be exerted when TD-

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containing serum interfered with lung cells BEAS-2B ($P < 0.01$), suggesting that the intestinal ameliorative effect of TD may be exerted indirectly through the lung.

Conclusion: TD can ameliorate NO–CO metabolism in the lung and thus achieve the indirectly amelioration of NO–CO metabolism in the intestine, ultimately achieving co-regulation of lung and intestinal inflammation, immune imbalance, cellular barrier damage, oxidative stress and intestinal bacterial disorders in asthma *in vivo* and *in vitro*. Targeting lung and intestinal NO–CO metabolic disorders in asthma may be a new therapeutic idea and strategy for asthma.

Abbreviations

CO	carbon dioxide
F	respiratory frequency per minute
FTTC	Fluorescein Isothiocyanate
FOXP3	Forkhead box protein 3
GT	gastrin
HO-1	heme oxygenase 1
IL-1 β	Interleukin-1 beta
HE	Haematoxylin-Eosin Staining
iNOS	Inducible Nitric Oxide Synthase
IL-4	Interleukin-4
IL-17	Interleukin-17
IFN- γ	Recombinant Human Interferon gamma
LEfSe	Linear discriminant analysis Effect Size
MTL	Motilin
Keap1	kelch-like ECH-associated protein 1

NO	nitric oxide
NF- κ B-P65	nuclear factor- κ B p65
M	model
NC	normal control
Nrf2	Nuclear Factor erythroid 2-Related Factor 2
OVA	ovalbumin
PCR	Polymerase Chain Reaction
Penh	hyperresponsiveness
P-NF- κ B-P65	Phospho nuclear factor- κ B p65
PCoA	Principal Co-ordinates Analysis
TCM	Traditional Chinese Medicine
TEER	transepithelial electrical resistance
TD	Tingli Dazao Xiefei Decoction
ROS	reactive oxygen species
WB	Western Blot
Y	Positive drugs

1. Introduction

Asthma is a chronic inflammatory disease that affects approximately 4.3% of the population worldwide (Khan et al., 2018; Guimarães et al., 2023) and is the most common chronic disease among children in the United States (Nakajima et al., 2012). By 2025, the agency estimates, there will still be nearly 100 million asthmatics (Tsabouri et al., 2017). In current scholarship, it is generally believed that the pathogenesis of asthma is due to airway inflammation and airway hyperresponsiveness due to chronic allergen irritation (Peebles and Aronica, 2019), and as the disease progresses, a series of irreversible structural and histological changes occur, ie, airway remodeling (Mims, 2015). Current clinical practice often uses glucocorticoids for this pathogenesis (Adcock and Mumby, 2017). The short time can significantly control the condition of patients with asthma. However, due to the recurring attacks of asthma, long-term use of glucocorticoids can lead to side effects such as gastro-duodenal ulcer, Cushing's syndrome, and steroid diabetes, which makes asthma a worldwide public health problem (Sweeney et al., 2016; He et al., 2019). Due to the reasons, we speculate that the current study have not fully mastered the specific pathogenesis of asthma. With the in-depth study on the pathogenesis of asthma in recent years, more and more researchers began to pay attention to some pathological changes in the intestine during the pathogenesis of asthma, and proposed the concept of "lung-intestine" axis (Tan et al., 2014; Sencio et al., 2021; Wang et al., 2022). The concept of "lung-intestine" axis has been described as early as in the Chinese medical book Inner Canon of Huangdi, which is widely used in the treatment of lung and intestinal diseases (Fu et al., 2018). Modern studies have also shown that there is a high degree of histological and pathological correlation between lung and intestine (Ni and Gao, 2012; Fu et al., 2016; Frati et al., 2018).

One of the high correlations between lung and intestine is mainly reflected in lung-intestinal gas exchange. Lung is the main organ for gas exchange of the body. Similarly, gas exchange of the intestine is also mainly discharged by lung (Bjørneklett and Jenssen, 1980). Two small

gas molecules, NO and CO, are widely present in the lung and intestine. Both are involved in the constitutive NO–CO metabolism which is closely related to the physiological and pathological functions of lung and intestine (Chung et al., 2008). NO is a kind of gas molecule free radical, chemically active. NO is mainly formed by nitric oxide synthase (NOS) catalyzed by L-arginine in the presence of reductive coenzyme II and oxygen (Aktan, 2004; Stojak et al., 2018). Studies have shown that NO is often used as a marker for airway inflammation (Ricciardolo, 2003), and its large expression can recruit, activate and participate in the activation of inflammatory NF- κ B signaling pathway to generate oxygen free radicals and aggravate airway inflammation (Wu et al., 2011; Oleson and Corbett, 2018). Studies have also reported that elevated NO is closely related to immune disorders (Ibiza et al., 2006), intestinal barrier damage (Wang et al., 2021) and intestinal microflora disorders (Hong et al., 2009). There are a number of drug studies related to NO. It has been demonstrated that Oridonin exerts an inhibitory effect by inhibiting NO expression exerting an improved anti-pneumonia and anti-fibrotic effect (Gao et al., 2022). Fisetin has also been shown to have an anti-pneumonia effect by inhibiting the expression of NO (Sim et al., 2020). Also NO has been shown to be a key target of the regulation for intestinal motility (Fichna et al., 2015).

CO is a small molecule gaseous substance with a structure similar to NO. CO is also an important messenger molecule, which is mainly derived from the oxidative decomposition of heme catalyzed by heme oxidase (HO) (Maines, 1988). CO can modify cellular processes by acting on specific intracellular proteases and indirectly interfering with signaling pathways such as NF- κ B signaling pathway and STAT3 signaling pathway (Fichna et al., 2015), thus producing significant anti-inflammatory, anti-apoptotic and anti-proliferative effects (Brouard et al., 2000; Kim et al., 2006; Ryter et al., 2006). It has been reported that exogenous administration of low concentrations of CO can significantly reduce the levels of eosinophils and mast cells in asthma model animals and exert significant anti-inflammatory effects (Chapman et al., 2001). Regarding CO, there are likewise a number of relevant drug studies. It has been shown that quercetin can upregulate CO-related protein HO-1 levels, which in turn exerts a protective effect on lung

epithelial cells (Hayashi et al., 2012). CO has found to be an important target for lung protection by interfering with related proteins (Mishra et al., 2006), and has also been shown to have a colonic mucosal repair effect *in vitro*, which is a vital target for colonic protection (Yoshinaga et al., 2004; Uchiyama et al., 2010). NO and CO will form a closed-loop NO–CO metabolic cycle (Hartsfield et al., 1997), that is, CO can feedback inhibit the increase of NO in pathological conditions (Thorup et al., 1999), so as to avoid excessive production of NO in pathological conditions and damage to cells or tissues. This indicates that the balance of NO and CO in the body is crucial for human lung and intestinal health whether it is physiological or pathological (Maulik et al., 1996; Caudill et al., 1998), and there are not many drugs studies for NO–CO metabolism consisting of NO and CO. Expanding drug studies on NO–CO metabolism is important to further to explore lung and intestinal related diseases such as asthma.

The Tingli Dazao Xiefei Decoction (TD) is a formula commonly used in China for the treatment of clinical lung diseases (Su and Xiong, 2021), and its formula consists of Tinglizi (*Descurainia sophia* (L.) Webb ex Prantl.) and Dazao (*Ziziphus jujube* Mill.), first recorded in Synopsis of the golden chamber by Zhongjing Zhang (Zhao et al., 2021). It has definite curative effect on asthma treatment (Zheng et al., 2014; Bo et al., 2018), and has the advantages of less side effects, high safety and good curative effect. The concept of traditional Chinese medicine treatment of diseases has always been the principle of comprehensiveness and integrity, such as “lung-intestine” axis. We speculate that TD, as a traditional Chinese medicine prescription, should also conform to the principle of overall medication of traditional Chinese medicine, that is, lung and intestine should be treated together. Further literature review shows that there are few studies on the effect of TD on NO–CO metabolism in the lung and intestine of asthma. Based on this, we query whether TD can simultaneously improve lung and intestinal injury caused by asthma? What are the mechanisms and potential ways for it to improve its effectiveness? To identify our questions, we select TD and asthmatic rats as study subjects in an attempt to clarify the ameliorative effects, mechanisms and potential pathways of TD on lung and intestinal NO–CO metabolism in asthma.

2. Materials and methods

2.1. Materials

2.1.1. Reagents

Grade II OVA (A5503, Sigma, USA); Grade V OVA (A5503, Sigma, USA); Aluminum hydroxide adjuvant (77161, Thermo, USA); DMEM (1898960, Gibco, USA); SDS-PAGE gel preparation kit (64136301, Bio-Rad, Hercules, CA, USA); Anti-IL-1 β (ab9722, Abcam, Cambridge, UK); Ficoll-Paque PREMIUM 1.084 sterile solution (17544602, cytiva, USA); Cell Stimulation Cocktail (plus protein transport inhibitors) (500X) (00-4975-93, Thermo, USA); eBioscience™ Foxp3 /Transcription Factor Staining Buffer Set (00-5523-00, Thermo, USA); Collagenase, Type 2 (WBC-LS004176, Worthington, USA); CD3 Monoclonal Antibody-PerCP-eFluor710 (46-0030-82, Thermo, USA); CD8a Monoclonal Antibody- PE-Cy7 (25-0084-82, Thermo, USA); CD4 Monoclonal Antibody-FITC (11-0040-82, Thermo, USA); IFN gamma Monoclonal Antibody-eFluor 660 (50-7310-80, Thermo, USA); IL-17A Monoclonal Antibody-PE-Cy7 (25-7177-82, Thermo, USA); CD25 Monoclonal Antibody- APC (17-0390-82, Thermo, USA); FoxP3 Monoclonal Antibody- PE (12-4774-42, Thermo, USA); PE Mouse Anti-Rat IL-4 (BD-555082, BD, USA); Anti-P-NF- κ B-P65 (AF2006, Affinify, China); Anti-NF- κ B-P65 (AF5006, Affinify, China); Anti-iNOS (AF0199, Affinify, China); Anti-Keap1 (AF5266, Affinify, China); Anti-Nrf2 (AF0639, Affinify, China); Anti- β -Tubulin (AF7011, Affinify, China); Anti-Claudin 5 (AF5216, Affinify, China); Anti-Connexin 43 (26980-1-AP, Proteintech, Wuhan, China); Anti-HO-1 (A1346, Abclonal, China); Cytoplasmic Membrane Extraction Kit (P0028, Beyotime, China); BeyoRTTM III First Strand cDNA Synthesis Kit (D7178M, Beyotime, China); BCA Protein Quantification

Kit (PC0020, Solarbio, Beijing, China); Fluorescein Sodium Salt (F8140, Solarbio, Beijing, China); Total RNA Extraction Kit (R1200, Solarbio, China); Reactive Oxygen Detection Kit (DCFH-DA) (CA1410, Solarbio, China); Nitric Oxide (NO) assay kit (A013-2-1, Nanjing Jiancheng Company, China); Endogenous carbon monoxide assay kit (A101-2-1, Nanjing Jiancheng, China); MTL ELISA Kit (E-EL-R0639c, Elabscience, Wuhan, China); GT ELISA Kit (E-EL-R0472c, Elabscience, Wuhan, China); D-xylose test kit (A035-1-1, Nanjing Jiancheng, China); ChamQ Universal SYBR qPCR Master Mix (L/N8E271E8, Vazyme, Nanjing, China); fetal bovine serum (Zhejiang Tianhang Biotechnology, Hangzhou, China); Trichloroacetaldehyde hydrate (Guoyao, Shanghai, China); Sodium carboxymethyl cellulose (Maikelin, Shanghai, China); other chemical reagents are all commercially available analytical grades.

2.1.2. Instruments

CO₂ incubator (Thermo, USA); Varioskan LUX Multifunctional Enzyme Labeler (Thermo, USA); BD FACS Aria III flow cytometer (BD, USA); Orthomorph microscope (Nikon, Shinagawa, Tokyo, Japan); Microplate reader (Bio-Tek, Winusky, VT, USA); Real-time fluorescence quantitative PCR instrument (Applied Biosystems, Waltham, MA, USA); Millicell-ERS-2 resistance system (Millicell-ERS; Millipore, MA, USA); Odyssey dual-color infrared fluorescence imaging system (Odyssey, USA); Mini-PROTEANTetra Electrophoresis tank (Bio-Rad, USA); TransBlot Turb System all-round protein transfer instrument (Bio-Rad, USA); AB204N precision analytical balance (Mettler To-ledo, Stockholm, Sweden); Rat airway resistance and lung compliance invasive detection system, (BUXCO, USA); Rat metabolic cage (Feng's Laboratory Animal, Jiangsu, China); Model 402A1 ultrasonic atomizer (Yuyue, Jiangsu, China); Biochemical incubator (Hengyi, Jiangsu, China); 0.4 μ m PET Transwell inserts (Labselect, Anhui, China); –80 °C ultra-low temperature refrigerator (Haier, Qingdao, China); –20 °C low temperature refrigerator (Haier, Qingdao, China).

2.2. In vivo method

2.2.1. Animals

7-week-old SPF male SD rats, weighing 180–200g, were purchased from Beijing Weitong Lihua Experimental Animal Technology Co., Ltd. The animal license number is SCXK (Beijing) 2016-0006. Artificial feeding was carried out in the standard animal laboratory at 18–22 °C. During the feeding period, free diet and drinking water were available. After one week of adaptive feeding, the experiment began. All animal procedures and protocols were approved by the Animal Ethics Committee of Henan University of Chinese Medicine. The ethical review approval number is DWLL2018080003.

2.2.2. Tingli Dazao Xiefei Decoction

The TD formula consists of Tinglizi and Dazao. Both Tinglizi and Dazao were provided and decocted by the herbal pharmacy of the Third Affiliated Hospital of Henan University of Traditional Chinese Medicine (Zhengzhou, Henan, China). The clinical human dose (15 g/kg), combined with the human and rat body surface area conversion factors, was converted to a rat equivalent medium dose (3.5 g/kg) as the dose of TD by gavage in rats. This dose is the best effective dose selected by the group after pre-screening and validation (B. Zhang et al., 2022). The samples were stored in The Engineering and Technology Center for Chinese Medicine Development of Henan Province, sample no. 20211121.

2.2.3. Grouping and administration

After the acclimatization feeding, the experimental animals were divided into, normal group (NC), model group (M), Tingli Dazao Xiefei Decoction group (TD, 3.5 g/kg), and dexamethasone group (Y, 0.075 mg/kg) (B. B. Zhang et al., 2022; B. B. Zhang et al., 2022) according to the principle of uniform body weight, 10 animals in each group.

Referring to previous studies (Elwood et al., 1991) and the results of our own pre-experiments, as shown in Fig. 1a, the experimental animals were modeled on two phases: the sensitization phase and the excitation phase (Xiong et al., 2007; Shaban et al., 2022). On days 1 and 8, except for the NC group, the rats in the other groups were sensitized by intraperitoneal injection of sensitizing solution (10 mg/mL grade V OVA + 10 mg/mL aluminum hydroxide) (1 mL/200 g), and the NC group was given an equal amount of saline intraperitoneally as a control. From day 15–21, groups M, TD and Y were nebulized for 7 consecutive days using the stimulating solution (2 g/100 mL grade II OVA) for 30 min each time, and the administered groups were administered by gavage with TD and dexamethasone for intervention, and groups M and NC were administered by gavage with equal amounts of distilled water as control.

The successful modeling was marked by frequent nodding breathing, shortness of breath, cyanosis of lips and coughing and wheezing.

2.2.4. Basic lung function

At the end of rat modeling, we measured airway hyperresponsiveness (Penh), respiratory frequency per minute (F), cough and wheeze to evaluate changes in basal lung function of asthmatic rats. The methods were as follows.

Rats were placed in a body surface tracing box and the Penh and F were measured by using the Rat airway resistance and lung compliance invasive detection system (Glaab et al., 2007; Glaab and Braun, 2021). Based on the previous studies and pre-experiments in our laboratory (Jia et al., 2022; N. Li et al., 2022), asthmatic rats were stimulated with 25%

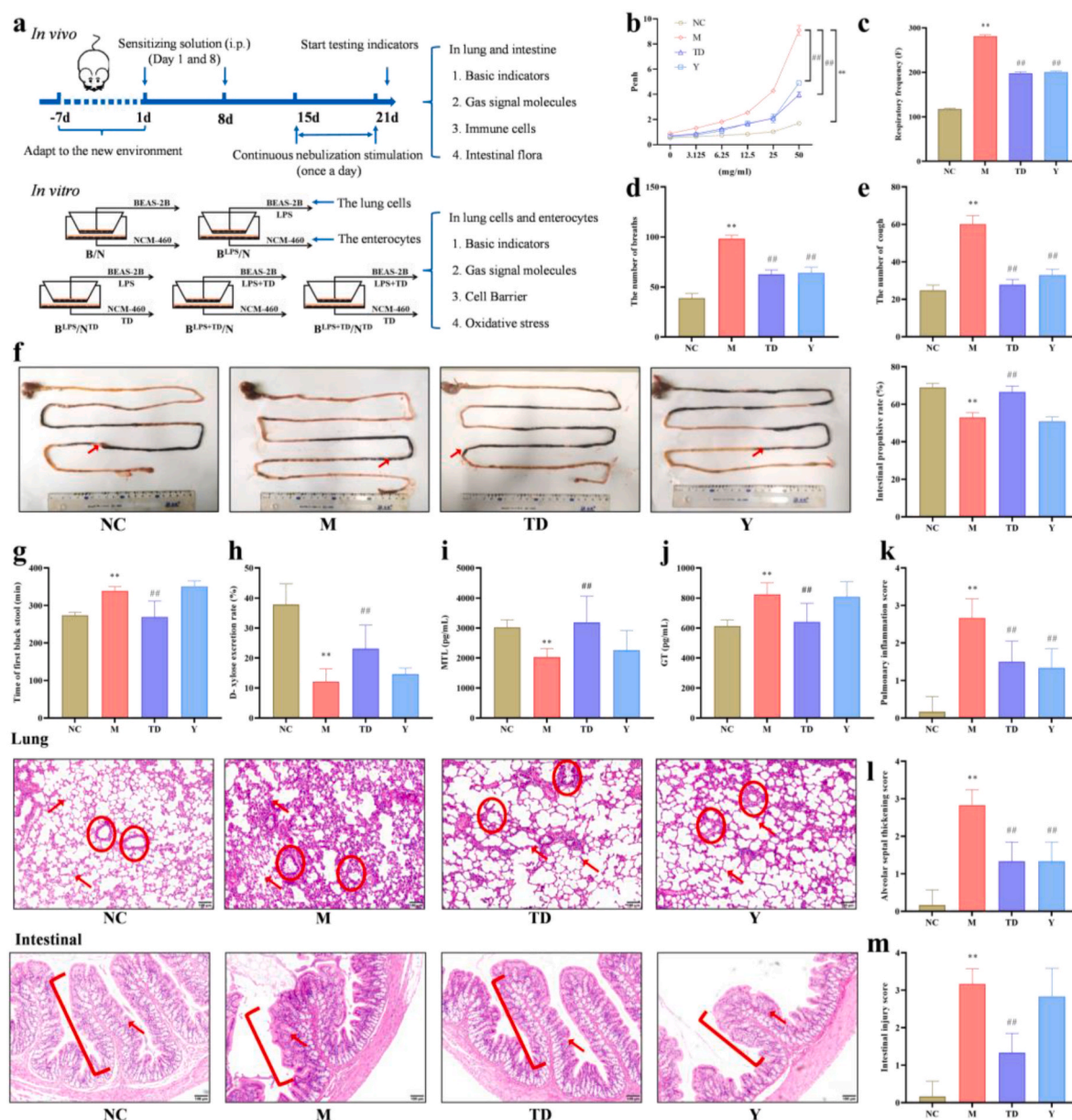


Fig. 1. Effect of TD on basal lung function and basal intestinal function in asthmatic rats. (a) Overview diagram of *in vivo* and *in vitro* experiments (b) Changes in penh in asthmatic rats (n = 6). (c) Changes in respiratory frequency in asthmatic rats (n = 6). (d) Changes in the number of breaths in asthmatic rats (n = 6). (e) Changes in the number of coughs in asthmatic rats (n = 6). **P < 0.01 compared with NC group, (f) Changes in small intestine propulsion rate in asthmatic rats (n = 6). (g) Variation in the timing of the first black stool discharge in asthmatic rats (n = 6). (h) Changes in the excretion rate of D-xylose in asthmatic rats (n = 6). (i) Changes in MTL levels in plasma in asthmatic rats (n = 6). (j) Changes in GT levels in plasma in asthmatic rats (n = 6). (k) Lung inflammatory cell infiltration scoring in asthmatic rats (n = 6). (l) Alveolar septal thickening scoring in asthmatic rats (n = 6). (m) Intestinal injury scoring in asthmatic rats (n = 6). ##P < 0.01 compared with M group. **P < 0.01 compared with NC group, ##P < 0.01 compared with M group.

ammonia to detect the number of coughs and 2% acetylcholine chloride and 0.1% histamine phosphate to detect the number of wheezes.

2.2.5. Basic intestinal function

To evaluate changes in basal intestinal function of asthmatic rats, we measured gastrointestinal hormone levels and evaluated basal gastrointestinal motility using D-xylose excretion rate, small intestinal propulsion rate and time to first black stool. The methods are as follows.

The levels of gastrointestinal hormones MTL and GT in rat serum were measured in strict accordance with the instructions of the MTL and GT ELISA kit.

Rats were allowed to drink and fast for 12 h. Rats were given 5% D-xylose solution by administered by gavage at 1 mL/100g, placed in a metabolic cage for 5h to collect urine and record urine volume. Later, the changes of intestinal absorption capacity of asthmatic rats were detected according to the instruction of D-xylose test kit.

The rats were administered by gavage with 1 mL/100g of a semi-solid nutritional paste containing 0.5% sodium carboxymethylcellulose in 10 g/L activated carbon. After 30min, the rats were rapidly anesthetized and the small intestine was taken out. The total length of the pylorus to the ileocecal part and the distance from the pylorus to the advancing front of the black semi-solid paste was measured, and the small intestine advancing rate was calculated and photographed.

The gavage method and dose were the same as those for measuring the small intestine advancement rate, and the rats in each group waited for the first black feces to be excreted, and the time spent was recorded.

2.2.6. Haematoxylin-Eosin staining (HE)

The right anterior lobe of the lung and intestine of each group of rats were uniformly excised for HE pathological examination. The tissues of each group were fixed by immersion in 4% paraformaldehyde for 24 h, embedded in paraffin, sectioned, and then stained for HE. Afterwards, the pathological and histological changes in the lung and intestinal tissues were observed under the microscope by two uninformed pathologists. lung inflammatory cell infiltration and alveolar septal thickening were scored with reference to literature methods (Underwood et al., 1995; Temelkovski et al., 1998), and the details of the scoring are shown in Table 1 and Table 2. Similarly, the colon was scored for intestinal tissue damage index with reference to the literature method (Jia et al., 2020), and the scoring details are shown in Table 3.

2.2.7. NO and CO detection

The lung and intestinal tissues were weighed and added to saline at a weight-to-volume ratio of 1:9 to make a 10% tissue homogenate. 3500 rpm was centrifuged and the supernatant was taken according to the instructions of the NO assay kit and the endogenous CO assay kit to detect changes in NO and CO in each group of lung and intestine.

2.2.8. Western blot (WB)

Appropriate amounts of lung and intestinal tissues were taken, and cell pulp and cell membrane proteins were extracted using a cytoplasmic membrane extraction kit, and protein quantification was performed using a BCA protein quantification kit. A total of 50 µg of cell plasma or cell membrane proteins were loaded per well for 10% SDS-PAGE gel electrophoresis, half-dry transferred to PVDF membrane, 5% BSA

Table 1
Lung inflammatory cell infiltration scoring criteria.

Evaluation criteria	Score
Normal (no or occasional inflammatory cell infiltration)	0
Mild inflammatory diffuse infiltration (rare inflammatory cell infiltration or occasional aggregation of inflammatory cells in clusters)	1
Moderate inflammatory infiltrate (inflammatory cells gathered in clusters, but did not invade the submucosa of the bronchus)	2
Severe cellular infiltration (a large number of inflammatory cells infiltrate in clusters and invade the bronchial submucosa or even the mucosal layer)	3

Table 2

Alveolar septal thickening scoring criteria.

Evaluation criteria	Score
Normal (no significant alveolar septal thickening)	0
Mild thickening (alveolar septal thickening to 1–2 times normal thickness)	1
Moderate thickening (alveolar septal thickening of 3–5 times normal thickness)	2
Severe thickening (alveolar septal thickening of more than 5 times the normal thickness)	3

Table 3

Intestinal injury scoring criteria.

Evaluation criteria	Score
Normal mucosa	0
Basal crypt defect 1/3	1
Basal crypt defect 2/3	2
Crypt absence, only the remaining surface epithelium, with inflammatory cell infiltration	3
Mucosal erosion, ulceration with a large number of inflammatory cell infiltration	4

closed, inflammation-related protein-specific antibodies (NF-κB-P65, P-NF-κB-P65, IL-1β), NO-CO metabolism-related antibodies (iNOS, HO-1) and β-tubulin were added, and after overnight at 4 °C, secondary antibodies were added and incubated at room temperature for 1 h. The membranes were washed and scanned using an Odyssey two-color infrared fluorescence imaging system, then the Image Studio software was used for statistical data analysis.

2.2.9. Flow cytometry detection of immunity

The lung and intestinal tissues were taken from the same location in each group of rats, and the lung and intestinal tissues were digested using 0.25% trypsin +0.01% type II collagenase and subjected to 70 µm cell straining to obtain lung and intestinal tissue cells, and then the obtained cells were purified using lymphocyte isolation solution. After 6 h of stimulation with cell stimulating agents, the numbers of CD4⁺T cells (CD3⁺ CD4⁺), CD8⁺T cells (CD3⁺ CD8⁺), Th1 cells (CD3⁺ CD4⁺ IFN-γ⁺), Th2 cells (CD3⁺ CD4⁺ IL-4⁺), Th17 cells (CD3⁺ CD4⁺ IL-17⁺) and Treg cells (CD3⁺ CD4⁺ CD25⁺ FoxP3⁺) were measured by flow cytometry to assess the changes of lung and intestinal immunity in asthmatic rats.

2.2.10. Intestinal flora

Each group of fresh feces was collected and PCR amplification of the V3 ~V4 region of the bacterial 16S rDNA gene fragment was commissioned from Zhejiang Hangzhou Lianchuan Biotechnology Co. The final clean data were clustered and sequences with similarity ≥97% were considered as one operational taxonomic unit (OTUs). After singleton filtering, the final OTU abundance and representative sequences were further analyzed by α-diversity analysis, β-diversity analysis, and LDA Effect Size (LEfSe) difference analysis.

2.3. In vitro method

2.3.1. Cell culture

Human normal lung epithelial cells (BEAS-2B cells) were purchased from Shanghai Enzyme Research Biotechnology Co., Ltd. and Human normal colonic epithelial cells (NCM-460 cells) were purchased from Beina Biotechnology Co. Both cells were cultured in high sugar DMEM containing 10% fetal bovine serum in a CO2 incubator at 5% CO2, 37 °C and saturated humidity, and cells at logarithmic growth stage were taken for passaging and experiments.

2.3.2. Preparation of TD drug-containing serum

The preparation and grouping methods of the drug-containing serum

were determined with reference to previous studies by scholars (Du et al., 2020; P. Zhou et al., 2022) and the results of our pre-experiments. SD rats were divided into four groups ($n = 5$): control group, TD-L group (1.75 g/kg TD), TD-M group (3.5 g/kg TD) and TD-H group (7 g/kg TD). The rats in the TD-containing serum group were given different doses of TD by gavage twice a day for 7 consecutive days, while the control group was given an equal amount of distilled water by gavage. One hour after the last gavage, the rats were anesthetized by intraperitoneal injection of 10% chloral hydrate, and the blood was collected from the abdominal aorta under aseptic conditions. The serum was inactivated in a water bath at 56 °C for 30 min and passed through a 0.22 µm filter membrane and stored at -20 °C until used for pharmacological studies.

2.3.3. Group of Co-culture

We constructed an *in vitro* cell model of lung injury causing intestinal injury by using 0.4 µm PET Transwell inserts. Briefly, we inoculated BEAS-2B cells at a density of 8×10^4 /mL in the upper chamber and NCM-460 cells at a density of 5×10^4 /mL in the lower chamber to enable the BEAS-2B/NCM-460 co-culture system to be established.

As shown in Fig. 1a, the cells were divided into BASE-2B/NCM-460 co-culture group (B/N group), BASE-2B^{LPS}/NCM-460 (B^{LPS}/N group), BASE-2B^{LPS}/NCM-460^{TD} (B^{LPS}/N^{TD} group), BASE-2B^{LPS} + TD/NCM-460 (B^{LPS} + TD/N group), BASE-2B^{LPS} + TD/NCM-460^{TD} (B^{LPS} + TD/N^{TD} group). In the B/N group, we co-cultured BASE-2B cells and NCM-460 cells directly without any other disposal. In the B^{LPS}/N group, BASE-2B cells were stimulated with 10 µg/L LPS for 24 h and then allowed to co-culture with NCM-460 cells for 6 h. The treatment protocol for the B^{LPS}/N^{TD} group was: the only difference compared to the B^{LPS}/N group was to allow co-culture of BASE-2B cells with NCM-460 cells pretreated with 10% TD serum for 24 h. BASE-2B cells pretreated with 10 µg/L LPS and 10% TD serum for 24 h were co-cultured with NCM-460 cells for 6 h as the B^{LPS} + TD/N group. In the last group, BASE-2B cells pretreated with 10 µg/L LPS and 10% TD serum for 24 h were co-cultured with NCM-460 cells pretreated with 10% TD serum for 24 h for 6 h as B^{LPS} + TD/N^{TD} group. After the co-culture of all groups, the changes of NCM-460 cells affected by LPS-induced BEAS-2B cell injury in all groups were examined.

2.3.4. NO and CO detection

After constructing the co-culture system, each co-culture group of NCM-460 cells was collected by trypsinization without EDTA, and the cells were resuspended by adding PMSF containing PBS. After ultrasonic cell fragmentation, changes in the levels of NO and CO in each group were detected according to the instructions of the NO assay kit and the endogenous CO assay kit.

2.3.5. WB

After constructing the co-culture system according to the method of 2.3.3, consistent with the method in 2.2.8, the cytoplasmic and membrane proteins of NCM-460 cells in each co-culture group were extracted, and each group of inflammation-related proteins (NF-κB-P65, P-NF-κB-P65, IL-1β), NO-CO metabolism-related proteins (iNOS, HO-1), cell barrier-related proteins (Connexin-43, Claudin-5), and antioxidant system-related proteins (Nrf2, Keap-1) were examined.

2.3.6. Transmembrane electrical resistance (TEER) and paracellular permeability assay

Cell transmembrane resistance and cell permeability were examined for each co-culture group to evaluate the effect of BASE-2B cells injury on the NCM-460 cells barrier (S. H. Liu et al., 2010; Wang et al., 2021). We switched the position of the upper and lower chambers of NCM-460 cells and BASE-2B cells according to 2.3.3 so that NCM-460 cells were located on the polyethylene terephthalate membrane of the co-culture chambers. Changes in transmembrane resistance of NCM-460 cells were measured by using the Millicell-ERS-2 resistance system. 10 µg/mL of sodium fluorescein was added to the top of the co-culture chamber,

the culture medium of the lower chamber was collected in a black opaque 96-well plate, and the fluorescence intensity of sodium fluorescein at 480 nm excitation wavelength and 520 nm emission wavelength was measured by a microplate reader to assess the change in epithelial permeability of NCM-460 cells (Gong et al., 2022; X. Zhou et al., 2022).

2.3.7. Flow cytometry for ROS detection

NCM-460 cells in each group of BEAS-2B/NCM-460 co-culture system were collected by trypsin digestion without EDTA, and incubated with 10 µmol/L DCFH-DA for 20 min in an incubator at 37 °C using the reactive oxygen species detection kit (DCFH-DA) according to the instructions of the kit. Then, the ROS levels of each group of NCM-460 cells were checked on a computer.

2.4. Statistical data analysis

The obtained data were statistically analyzed by SPSS 20.0 software, and the data were expressed as mean ± standard deviation ($\bar{x} \pm s$). One-way analysis of variance was used to find the difference between the two groups. $P < 0.05$ indicated statistically significant differences between the groups, and $P < 0.01$ indicated extremely significant differences.

3. Results

3.1. Effect of TD on basal lung and intestinal function in asthmatic rats

After we replicated and grouped the rats in the asthma model by going through the process as described in 2.2.2. We evaluated the basal lung function of the rats in each group. The results showed that the penh index were significantly higher in the rats of group M compared to group NC ($P < 0.01$), which indicated that the rats of group M exhibited typical asthmatic airway hyperresponsiveness change (Fig. 1b). We also observed a significant increase in the respiratory frequency, number of coughs, and number of wheezes in the rats of the M group ($P < 0.01$) (Fig. 1c, d, e). All these phenomena indicated that the rats in the M group received severe impairment of basal lung function and showed typical symptoms of asthma, indicating that we successfully replicated the asthma model. Compared with the M group, these symptoms were alleviated in the TD and Y groups ($P < 0.01$), suggesting that both TD and dexamethasone could have restorative effects on the impaired lung function caused by asthma.

Several papers exist that have reported asthma-induced changes in intestinal function (He et al., 2021). We performed a more systematic assessment of gastrointestinal function in asthmatic rats to investigate in depth whether TD could improve the possible asthma-induced impairment of gastrointestinal function. Our results show that compared to group NC, group M showed significantly lower rates of small intestinal propulsion and D-xylose excretion and significantly higher time to first black stool ($P < 0.01$); in addition, we detected abnormal dysregulation of plasma levels of the common gastrointestinal hormones MTL and GT ($P < 0.01$) (Fig. 1f, g, h, i, j). The rate of D-xylose excretion is proportional to the level of intestinal absorption, which can reflect the strength of gastrointestinal function to some extent (Gao et al., 2009). The gastrointestinal hormones MTL and GT are closely related to gastrointestinal dynamics and transport in the gastrointestinal tract (Liu et al., 2020). These data suggest that there are multiple levels of impairment of intestinal function of asthmatic rats. These conditions showed remission and improvement after the administration of TD. This result suggests that the improvement and recovery of asthma by TD is multi-faceted and multi-dimensional, targeting both lung and intestine.

3.2. Effects of TD on lung and intestinal pathology in asthmatic rats

HE staining is the most recognized and commonly used staining method to observe histopathological histology. We evaluated the

interventional effect of TD on the lung and intestine of asthmatic rats by using HE staining. As shown in Fig. 1k, l, m, in the lung pathological sections of each group, we can clearly see that the normal group of rats had intact lung tissue structure and no obvious inflammatory cell infiltration or aggregation was seen, while asthmatic rats were seen to have obvious peribronchial massive inflammatory cell infiltration and damaged alveolar walls. In the pathological sections of the intestine, the colonic villi of normal rats were extended and no obvious inflammatory cell infiltration was seen, while the asthmatic rats obviously showed shortened colonic tissue intestinal villi and inflammatory cell infiltration around them. We further performed a systematic scoring of the lung and intestinal pathological sections in each group of rats. The results showed that the lung tissue inflammatory cell infiltration score and alveolar septal thickening score appeared significantly higher in the M group compared with the NC group ($P < 0.01$), and the intestinal tissue damage index score also increased simultaneously ($P < 0.01$). Lung and intestinal injury were significantly improved in the TD group compared to the M group ($P < 0.01$), and lung injury was significantly improved in the Y group ($P < 0.01$). This is a strong histopathological validation of the high correlation between lung and intestinal tissues, indicating that asthmatic rats cause not only lung injury but also intestinal injury to some extent. TD has a simultaneous histopathological improvement on lung and intestine.

3.3. Effects of TD on lung and intestinal NO–CO metabolism in asthmatic rats

The behavior of gas exchange of the lung and intestine is highly correlated, and NO and CO are two widely present gas signaling molecules in the organism. Reviewing the literature, there is little discussion of their co-alteration in asthma. Deeper digging into the literature revealed that iNOS protein catalyzes the formation of NO, HO-1 protein catalyzes the formation of CO, CO inhibits the expression of iNOS

protein, and NO has an inhibitory effect on HO-1 protein, and this interlocking relationship constitutes the balance of NO–CO metabolism (Hartsfield et al., 1997; Thorup et al., 1999). Whether the altered intestinal function and even damage caused by asthma are related to the balance between NO and CO and whether TD can be regulated bidirectionally to balance NO and CO? We focused on this and evaluated both the lung and the intestine. We measured NO and CO levels in lung and intestinal tissue homogenates using kits and detected changes in the relative levels of iNOS and HO-1 protein, which are closely associated with NO and CO, using the western bolt method. The results of Fig. 2a, b, c clearly reflect the changes of NO–CO metabolism in asthmatic rats. We can see that the levels of NO in the lung and intestine of the M group are significantly upregulated compared to the NC group ($P < 0.01$), while the levels of CO are significantly decreased ($P < 0.01$). Similarly, iNOS and HO-1 protein showed an increase and decrease ($P < 0.01$), respectively. Only lung and intestinal NO and iNOS showed highly significant regression after dexamethasone administration ($P < 0.01$), and all of these indicators showed highly significant regression after TD administration ($P < 0.01$). This result strongly demonstrates that there is a serious disturbance of NO–CO metabolism in the asthmatic rat organism, and that the improvement of this metabolism in the lung and intestine by TD is simultaneous, with the improvement of dexamethasone targeting only NO levels and iNOS protein levels. The lung and intestine of asthmatic rats show a disturbance in NO–CO metabolism, and this disturbance leads to elevated NO levels, and elevated NO usually brings about a severe inflammatory response (Warke et al., 2002). Disturbances in NO–CO metabolism lead to elevated NO levels, and NO is often used as a marker for airway inflammation (Ricciardolo, 2003). To deeply to evaluate the improvement pathway of TD on the lung intestine of asthmatic rats, we examined the inflammatory pathway-related proteins P–NF- κ B–P65 and IL-1 β by western bolt method. As shown in Fig. 2d and e, the relative expression levels of inflammatory pathway-associated proteins P–NF- κ B–P65 and IL-1 β were highly significantly upregulated

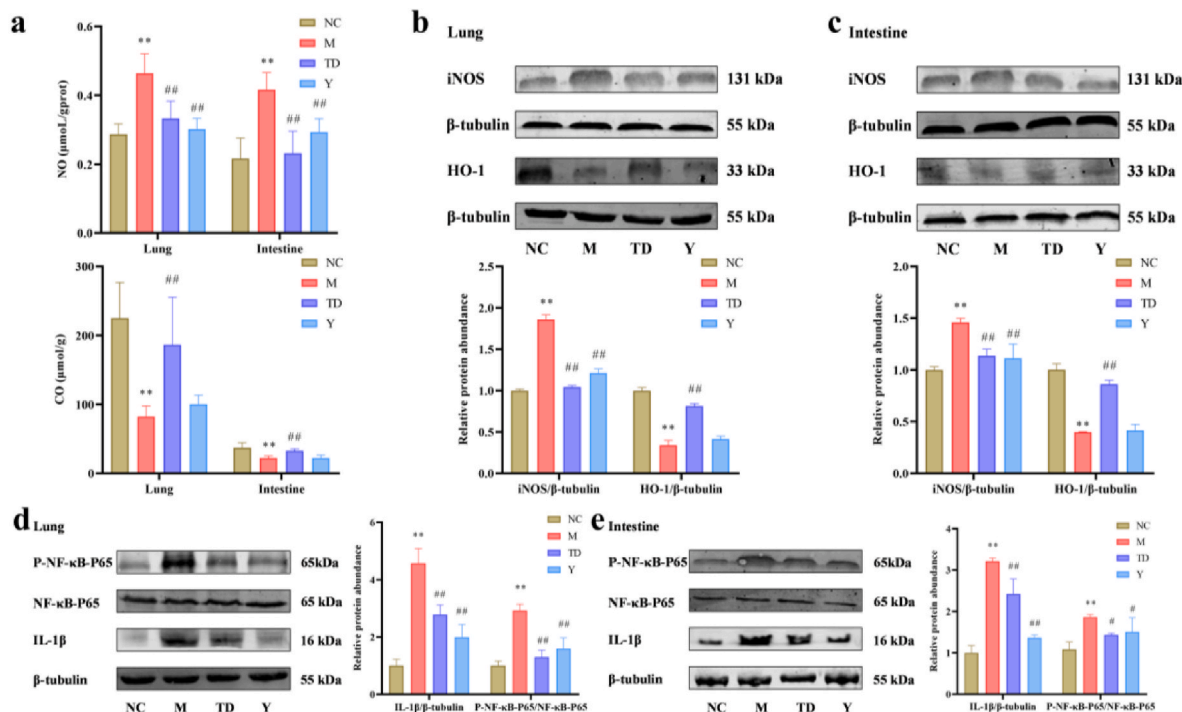


Fig. 2. Effects of TD on lung and intestinal NO–CO metabolism in asthmatic rats. (a) Changes in NO and CO in the lung and intestine of asthmatic rats ($n = 6$). (b) Representative images and quantification of changes in the relative levels of iNOS and HO-1 in the lung of asthmatic rats detected by western bolt ($n = 3$). (c) Representative images and quantification of changes in the relative levels of iNOS and HO-1 in the intestines of asthmatic rats detected by western bolt ($n = 3$). (d) Representative images and quantification of changes in the relative levels of P–NF- κ B–P65 and IL-1 β in the lung of asthmatic rats detected by western bolt ($n = 3$). (e) Representative images and quantification of changes in the relative levels of P–NF- κ B–P65 and IL-1 β in the intestines of asthmatic rats detected by western bolt ($n = 3$). ** $P < 0.01$ compared with NC group, ## $P < 0.01$ compared with M group.

in the M group rats compared with the NC group ($P < 0.01$). A significant down-regulation of the relative expression of inflammatory pathway-related proteins occurred after the administration of TD or dexamethasone ($P < 0.05$ or $P < 0.01$). This suggests that asthmatic rats have significant inflammation present in both lung and intestine, and both TD and dexamethasone can reduce lung and intestinal inflammation in asthmatic rats to some extent.

3.4. Effects of TD on lung and intestinal immune cells in asthmatic rats

The high production of NO is also closely related to immunity, which corresponds to the deregulated ratio of lung immune cells in asthma reported in the extensive literature, so based on our previous results, there is also a significant elevation of NO in the intestine of asthmatic rats, so is intestinal immunity disturbed by the crosstalk of NO–CO metabolism, and does TD has an ameliorating effect? We performed a systematic assay to evaluate asthmatic lung and intestinal immune cells (Fig. 3). The results showed a significant proportional imbalance in the CD4⁺T cells/CD8⁺T cells, Th1 cells/Th2 cells and Th17 cells/Treg cells ratio in the M group compared to the NC group ($P < 0.05$ or $P < 0.01$). TD had a significant ability to back-regulate the imbalance in immune cell ratios both in the lung and intestine compared to the M group ($P < 0.05$ or $P < 0.01$); whereas dexamethasone only had the ability to correct the imbalance in lung immune cells ($P < 0.05$ or $P < 0.01$). This is a very interesting result, which strongly suggests that the therapeutic effect of TD in asthma is a multidimensional improvement targeting the lung and intestine. Whereas dexamethasone appears to target only the lung, we speculate that this may have relevance to the other main protagonist of NO–CO metabolism, CO, as shown in Fig. 2a, b, c, where dexamethasone has no ameliorative effect on either the lung or the intestine in asthma ($P > 0.05$).

3.5. Establishment and evaluation of BEAS-2B /NCM-460 co-culture system

Both animals and humans are a whole, and *in vivo* lung-intestine interactions always involve other organs or mediators, although previous studies have indirectly demonstrated a lung-intestine correlation. However, we believe that there is still a lack of a direct evidence to prove the direct correlation between lung and intestine effects. Based on this we used human lung bronchial epithelium (BEAS-2B cells) and human normal colonic mucosal epithelial cells (NCM-460 cells) with the aim of establishing a lung-intestine co-culture system. The methodology for this part of the experiment is detailed in Supplemental methods. The inflammatory damage of LPS on BEAS-2B cells was able to pass on to NCM-460 cells co-cultured with them, causing their inflammatory factor levels (TNF- α , IL-1 α , IL-6) to peak at the 6th hour (Fig. S6). Based on this, we chose 6 h as the co-culture time for both cells. Normal growth of BEAS-2B cells did not affect NCM-460 cells ($P > 0.05$), and LPS stimulation of BEAS-2B cells either in the upper or lower compartment caused activation of NF- κ B inflammatory pathway and disruption of NO–CO metabolism in NCM-460 cells ($P < 0.01$) (Fig. 4a, b, c, d), which is consistent with the changes seen in animal experiments. This suggests that even if the upper and lower positions of the two cells in the co-culture system are different, it does not affect the transmission of damage from BEAS-2B cells to NCM-460 cells in the pathological state. Based on this we clarify that the BEAS-2B /NCM-460 co-culture system is feasible. After clarifying the feasibility of the BEAS-2B /NCM-460 co-culture system. We first performed a small experiment with LPS stimulation of BEAS-2B cells and NCM-460 cells alone and administered TD's drug-containing serum intervention. In this part of the experiment, we first determined the stimulation concentration of LPS in BEAS-2B cells and NCM-460 cells inflammatory injury model and excluded the cytotoxicity of TD's drug-containing serum (Fig. S3 and Fig. S4). Then we used different doses of TD's drug-containing serum to interfere with the inflammatory damage caused by LPS in BEAS-2B cells and NCM-460

cells, and the results of this part showed that different doses of TD's drug-containing serum ameliorated the LPS-induced elevation of inflammatory factors (TNF- α , IL-1 α , IL-6) in BEAS-2B cells ($P < 0.01$), but not in LPS-induced NCM-460 cells ($P > 0.05$), with the best effect of medium dose of TD's drug-containing serum in inflammatory injury in BEAS-2B cells (Fig. S5). Such experimental results led us to the conjecture that the ameliorative effect of TD on NCM-460 cells, the intestinal cells, requires the involvement of lung cells BEAS-2B cells, and by extension, in animals, that the ameliorative effect of TD on the intestine is indirectly regulated through the lung.

3.6. Effect of TD's drug-containing serum on NO–CO metabolism in BEAS-2B /NCM-460 co-culture system

To test our conjecture about the mode of action of TD amelioration, We selected the optimal dose of TD-containing serum and the optimal LPS-modeling concentration validated in the previous part of the screening for different treatments of BEAS-2B cells and NCM-460 cells in the co-culture system, and then detected changes in the classical NF- κ B inflammatory pathway to explore the mode and mechanism of TD action on BEAS-2B cells and NCM-460 cells, respectively. The results of this part of the experiment are shown in Fig. 5a, where we can observe that BEAS-2B stimulated by LPS led to the activation of NF- κ B signaling pathway in the lower chamber of NCM-460 cells ($P < 0.01$), and the administration of TD-containing serum to NCM-460 cells did not improve this condition ($P > 0.05$). The activation of NF- κ B signaling pathway in NCM-460 cells was inhibited only by administration of TD-containing serum to BEAS-2B cells or by simultaneous administration of TD-containing serum to BEAS-2B cells and NCM-460 cells ($P < 0.01$), and there was no significant difference between the two modalities ($P > 0.05$), i.e., there was no synergistic potentiation of the amelioration of TD by the two cells co-cultures. This verifies our conjecture that TD can regulate the inflammation of lung cells BEAS-2B to improve the inflammation of intestinal cells NCM-460.

The NO–CO metabolism-related indexes of the BEAS-2B /NCM-460 co-culture system were further examined. The results showed that after LPS stimulation, BEAS-2B cells in the upper chamber showed a significant increase in NO levels and a significant decrease in CO levels ($P < 0.01$), which indicated that LPS stimulation led to the disruption of NO–CO metabolism in BEAS-2B cells (Figs. S7D and E). We then examined the NO–CO metabolism-related indexes in NCM-460 cells located in the lower chamber, and the results as shown in Fig. 5b, c, d showed that LPS stimulation of BEAS-2B cells caused a significant increase in NO levels and relative expression of iNOS, and a significant decrease in CO levels and relative expression of HO-1 in NCM-460 cells in the lower chamber ($P < 0.01$). When we administered TD-containing serum to BEAS-2B cells or TD-containing serum to both BEAS-2B cells and NCM-460 cells, the NO–CO metabolic disorder in NCM-460 cells was significantly improved ($P < 0.01$). In contrast, the administration of TD-containing serum to NCM-460 cells alone did not exert an ameliorative effect ($P > 0.05$). This indicates that the improvement effect of TD requires the participation of lung cells BEAS-2B cells, which means that TD can improve the NO–CO metabolism disorder of intestinal cells NCM-460 by regulating the NO–CO metabolism disorder of lung cells BEAS-2B.

3.7. Effect of TD's drug-containing serum on the cell barrier and antioxidant system in BEAS-2B /NCM-460 co-culture system

The intestinal mucosa is composed of epithelial cells, which are an important barrier with defense functions. The health of the intestinal barrier is closely related to the permeability of the epithelial cells, which means that the permeability of the intestinal epithelial cells is important to the normal function of the intestine. So, in our BEAS-2B /NCM-460 co-culture system, does LPS-induced damage to lung cells BEAS-2B cells affect the cell permeability of intestinal epithelial cells NCM-460 cells

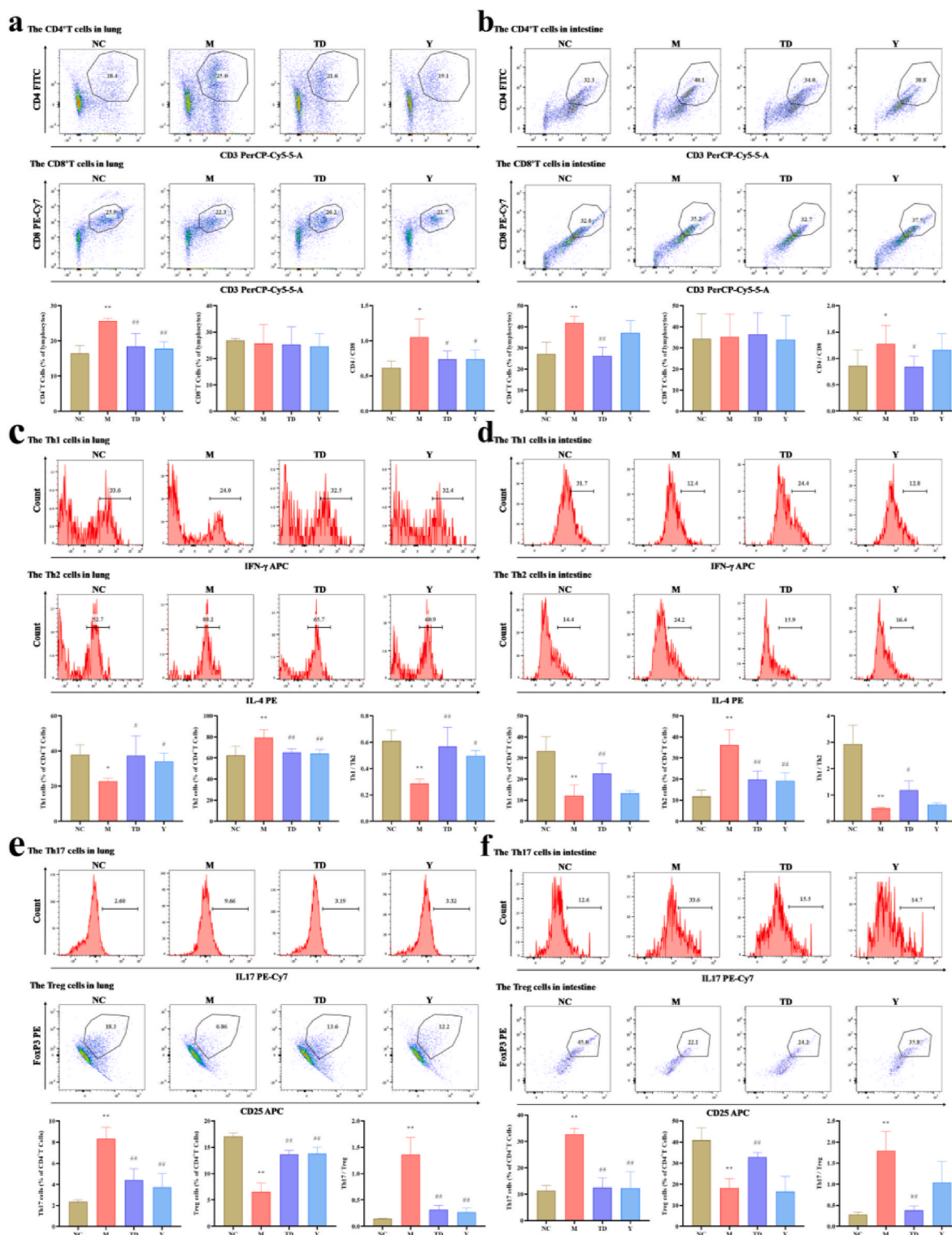


Fig. 3. Effects of TD on lung and intestinal immune cells in asthmatic rats. (a) Representative images and statistics change of CD4⁺T cells and CD8⁺T cells flow cytometry in lung of asthmatic rats (n = 6). (b) Representative images and statistics change of CD4⁺T cells and CD8⁺T cells flow cytometry in intestinal of asthmatic rats (n = 6). (c) Representative images and statistics change of Th1 cells and Th2 cells flow cytometry in lung of asthmatic rats (n = 6). (d) Representative images and statistics change of Th1 cells and Th2 cells flow cytometry in intestinal of asthmatic rats (n = 6). (e) Representative images and statistics change of Th17 cells and Treg cells flow cytometry in lung of asthmatic rats (n = 6). (f) Representative images and statistics change of Th17 cells and Treg cells flow cytometry in intestinal (n = 6). ***P* < 0.01 compared with NC group, #*P* < 0.05 compared with M group, ##*P* < 0.01 compared with M group.

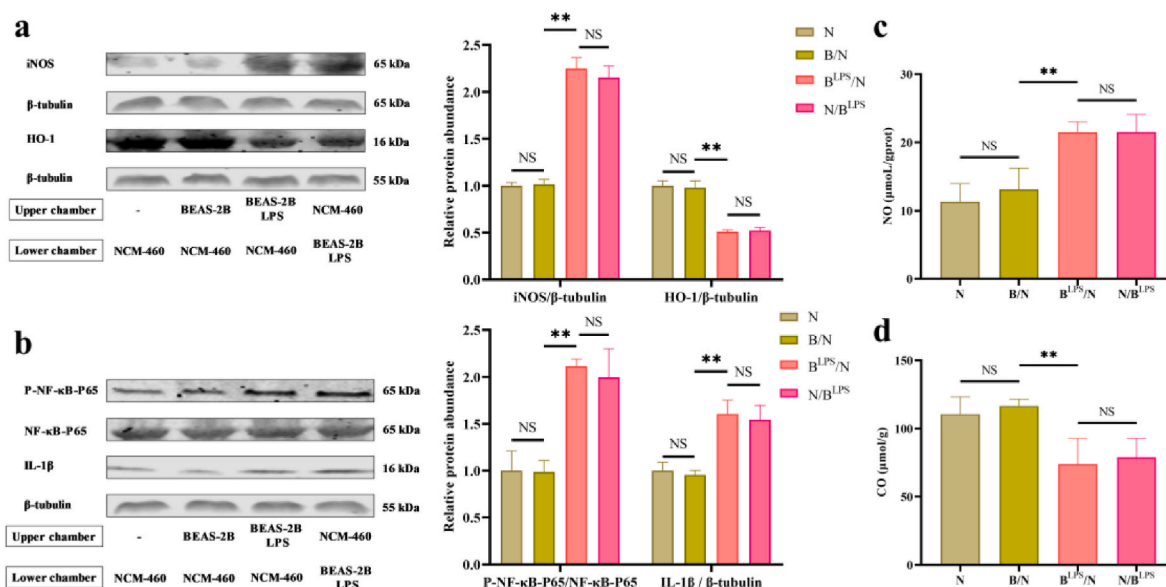


Fig. 4. Establishment and evaluation of BEAS-2B /NCM-460 co-culture system. (a) Representative images and quantification of the effect of upper and lower chamber interchange on the relative levels of iNOS and HO-1 proteins in NCM-460 cells of the BEAS-2B/NCM-460 co-culture system were examined by Western bolt (n = 3). (b) Representative images and quantification of the effect of upper and lower chamber interchange on the relative levels of P-NF-κB-P65 and IL-1β proteins in NCM-460 cells of the BEAS-2B/NCM-460 co-culture system were examined by Western bolt (n = 3). (c, d) The quantification of the effect of upper and lower chamber interchange on the relative levels of NO and CO in NCM-460 cells of the BEAS-2B/NCM-460 co-culture system were examined by kit (n = 6). Compared between the two groups, ** $P < 0.01$.

and what is the efficacy and manner of TD intervention? Based on this, we examined the changes in permeability in the BEAS-2B /NCM-460 co-culture system. The results showed that the transmembrane resistance of BEAS-2B cells were significantly reduced and the sodium fluorescein transmission rate of BEAS-2B cells were significantly increased after LPS stimulation ($P < 0.01$) (Figs. S5B and C), which indicated that LPS stimulation led to cell barrier damage in BEAS-2B cells. NCM-460 cells in the BEAS-2B/NCM-460 co-culture system were examined simultaneously, and the results as shown in Fig. 5e, f, g showed that the damage caused by LPS to BEAS-2B cells was transmitted to NCM-460 cells, and NCM-460 cells were affected by the cell barrier damage caused by BEAS-2B cells. NCM-460 cells also showed a significant decrease in transmembrane resistance and a significant increase in sodium fluorescein transmission rate ($P < 0.01$). The normal expression of cell gap junction proteins is the basis and key to cell barrier function. Connexin-43 and Claudin-5 are very common proteins between cell interstitial spaces and their normal expression is related to the homeostasis of the cell barrier. In-depth examination of their changes showed that the relative expression of Connexin-43 and Claudin-5 was significantly decreased in NCM-460 cells ($P < 0.01$). Cell barrier damage in NCM-460 cells was not improved when we treated NCM-460 cells alone with administration of TD-containing serum ($P > 0.05$). When BEAS-2B cells or both the BEAS-2B cells and NCM-460 cells were given TD-containing serum, the cell barrier damage of NCM-460 cells was significantly improved ($P < 0.01$). This suggests that the involvement of lung cells BEAS-2B is required when TD is to exert an ameliorative effect on intestinal cells NCM-460. TD can improve the cell barrier damage of intestinal cells NCM-460 by improving the cell barrier damage of lung cells BEAS-2B.

Damage to the intestinal barrier leads to increased cell permeability, and increased cell permeability leads to further oxidative stress. This creates a vicious cycle that leads to more severe damage to the intestinal barrier. Based on this, we deeply examined the changes of oxidative stress damage in BEAS-2B /NCM-460 co-culture system. We first evaluated the effect of LPS on lung cells BEAS-2B, and the results showed that BEAS-2B cells exhibited significantly elevated ROS levels after LPS stimulation ($P < 0.01$) (Fig. S5A). This suggests that LPS stimulation can lead to significant oxidative stress damage in BEAS-2B cells. Next, NCM-

460 cells in the BEAS-2B/NCM-460 co-culture system were examined, and the results as shown in Fig. 5h and i, showed that the oxidative stress damage caused by LPS to BEAS-2B cells was transmitted to NCM-460 cells, and NCM-460 cells showed a significant increase in ROS levels ($P < 0.01$). Keap1 and Nrf2 together constitute the main sensors of the cellular response to oxidative stress (Itoh et al., 1999; Ruan et al., 2021). To investigate the changes in oxidative stress levels in NCM-460 cells in depth, we examined the changes in the relative expression levels of Keap1 and Nrf2, and the results showed that the relative expression levels of Nrf2 protein decreased significantly and the relative expression of Keap-1 increased significantly ($P < 0.01$). This indicates that NCM-460 cells were severely affected by BEAS-2B cells and their anti-oxidant system was severely damaged. When we intervened with TD-containing serum, the pattern of its ameliorative effect on lung and intestinal cells was consistent with the previous part of the experiment, where TD exerted an ameliorative effect on oxidative stress in intestinal cells NCM-460 only in the presence of lung cells BEAS-2B ($P < 0.01$). This suggests that the ameliorative effect of TD on oxidative stress in intestinal cells NCM-460 requires the participation of lung cells BEAS-2B. TD can ameliorate oxidative stress damage of intestinal cells NCM-460 by ameliorating oxidative stress damage of lung cells BEAS-2B.

3.8. Effect of TD on intestinal flora of asthmatic rats

Many diseases are accompanied by changes in intestinal flora (Deng et al., 2022; Wen et al., 2022), and we also explored the effects of TD on intestinal flora in asthmatic rats. The chao1 index based on alpha diversity analysis showed that the chao1 index were highly significantly downregulated in the M group compared with the NC group, and both were significantly improved by the administration of TD ($P < 0.01$) (Fig. 6a). The PCoA results based on Bate diversity analysis can clearly show the clustering of the groups, and this part of the results shows that there is a significant trend of separation between the NC and M groups, a significant trend of separation between the TD and M groups, and a significant trend of proximity between the TD and NC groups (Fig. 6b). Then, LefSe analysis was performed on NC group and M group with

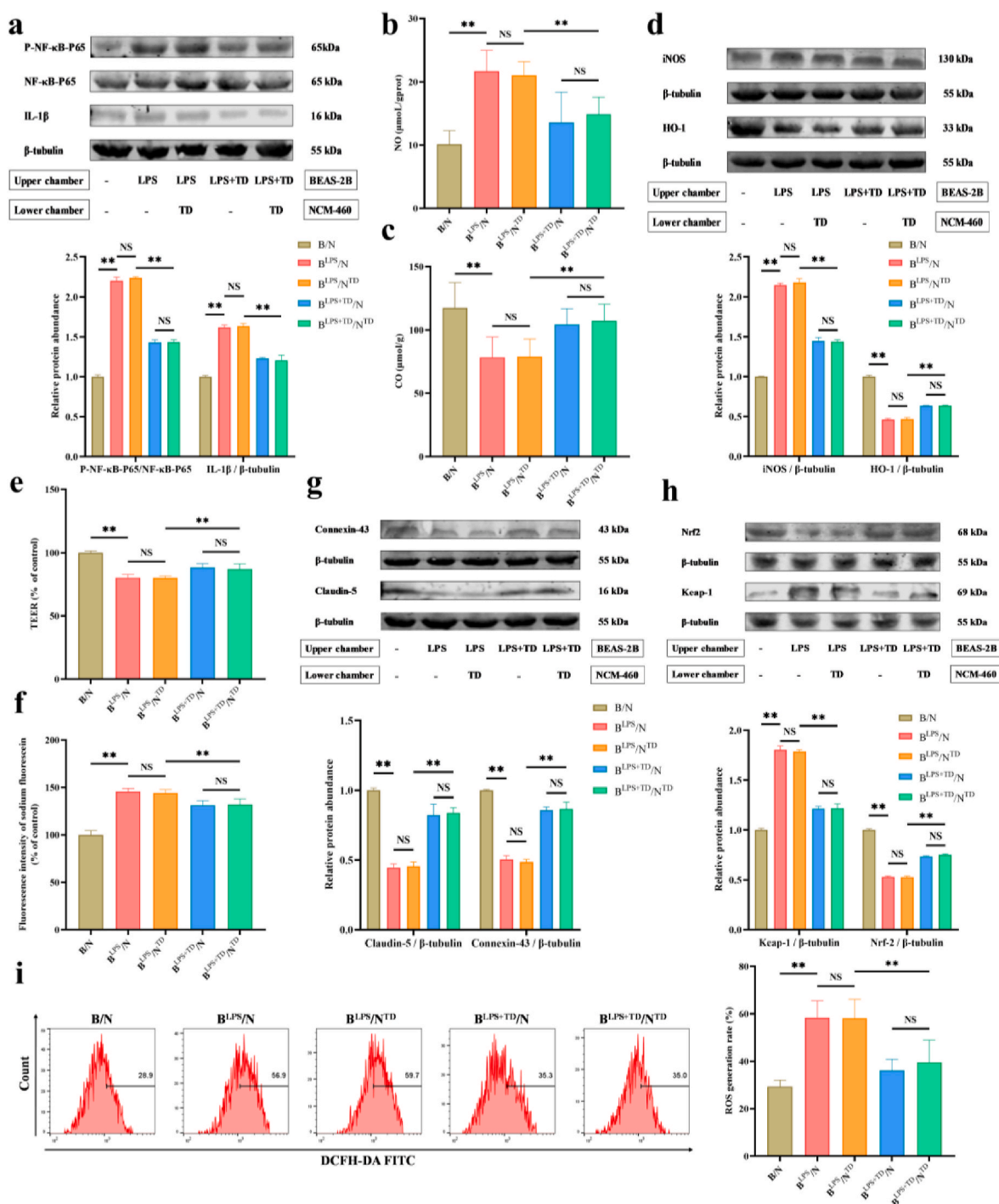


Fig. 5. Effect of TD's drug-containing serum on BEAS-2B /NCM-460 co-culture system. (a) Representative images and quantification of the effect of different modalities of TD's drug-containing serum treatment on the relative levels of P-NF- κ B-P65 and IL-1 β proteins in NCM-460 cells of the BEAS-2B/NCM-460 co-culture system were examined by Western bolt (n = 3). (b, c) The quantification of the effect of different modalities of TD's drug-containing serum treatment on the relative levels of NO and CO in NCM-460 cells of the BEAS-2B/NCM-460 co-culture system were examined by kit (n = 6). (d) Representative images and quantification of the effect of different modalities of TD's drug-containing serum treatment on the relative levels of iNOS and HO-1 proteins in NCM-460 cells of the BEAS-2B/NCM-460 co-culture system were examined by Western bolt (n = 3). (e) Effect of different modes of TD-containing serum treatment on TEER of NCM-460 cells in BEAS-2B/NCM-460 co-culture system (n = 6). (f) Effect of different modes of TD-containing serum treatment on sodium fluorescein leakage of NCM-460 cells in BEAS-2B/NCM-460 co-culture system (n = 6). (g) Representative images and quantification of the effect of different modalities of TD's drug-containing serum treatment on the relative levels of Connexin-43 and Claudin-5 proteins in NCM-460 cells of the BEAS-2B/NCM-460 co-culture system were examined by Western bolt (n = 3). (h) Representative images and quantification of the effect of different modalities of TD's drug-containing serum treatment on the relative levels of Keap1 and Nrf2 proteins in NCM-460 cells of the BEAS-2B/NCM-460 co-culture system were examined by Western bolt (n = 3). (i) Representative images and quantification of the effect of different modes of TD-containing serum treatment on ROS of NCM-460 cells in BEAS-2B/NCM-460 co-culture system (n = 6). Compared between the two groups, $^{**}P < 0.01$.

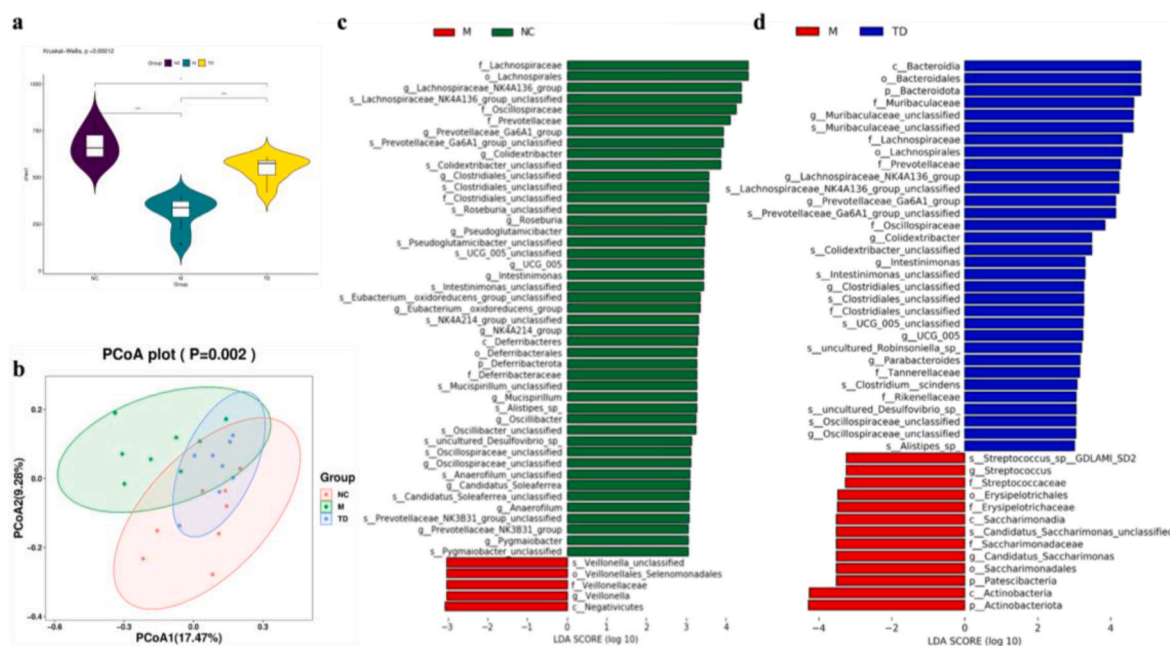


Fig. 6. Effect of TD on intestinal flora of asthmatic rats. (a) Results of change in chao1 index in asthmatic rats based on alpha diversity analysis ($n = 8$). (b) Results of change in PCoA analysis in asthmatic rats based on beta diversity analysis ($n = 8$). (c) Results of differential flora of NC and M groups based on LEfSe analysis ($n = 8$). (d) Results of differential flora of M and TD groups based on LEfSe analysis ($n = 8$). Compared between the two groups, *** $P < 0.001$, ** $P < 0.01$, * $P < 0.05$.

$LDA \geq 3.0$, as well as M group and TD group, to find the changes of the difference microbiota between the two groups after TD administration.

Data from LEfSe analysis showed that at the genus level, rats in the NC group shared changes with the M group and rats in the M group with the TD group in the *g_Lachnospiraceae_NK4A136*, *g_Prevotellaceae_Ga6A1*, *g_UCG_005*, *g_Intestinimonas* and *g_Colidextribacter*. A review of the literature revealed that *g_Lachnospiraceae_NK4A136* has anti-inflammatory and intestinal mucosal repair effects and can effectively alleviate intestinal inflammation (Zhong et al., 2020), *g_Prevotellaceae_Ga6A1* has a role in maintaining host immune homeostasis and intestinal mucosal repair (Guo et al., 2016; Zhu et al., 2020), *g_UCG_005* is associated with digestive function (Vojinovic et al., 2019), *g_Intestinimonas* and *g_Colidextribacter* are associated with immune dysregulation (Mager et al., 2020; P. Y. Li et al., 2022) (Fig. 6c and d). All the results from this part point in a clear conclusion that the TD has a significant modification effect on the intestinal flora of asthmatic rats.

4. Discussion

Asthma is a respiratory disease that affects about 334 million people worldwide and is typically characterized by recurrent airway inflammation (Nunes et al., 2017; Cerqua et al., 2022). Currently, there are no specific drugs for asthma treatment, and most of them use glucocorticoids and anticholinergic drugs, etc (He et al., 2019). However, due to the recurrent nature of asthma, the long-term use of asthma may lead to side effects of various drugs (Sweeney et al., 2016). Chinese medicine has a long history of treating asthma, and the combined treatment of lung and intestine is the consistent view of Chinese medicine on the treatment of lung diseases (Fu et al., 2018; Cai et al., 2023). Nowadays, more and more evidence has shown that the treatment of asthma cannot be limited to focus only on the changes in the lung, but also the changes in the intestines are key to the development of the course of asthma (Tan et al., 2014; Sencio et al., 2021). This fully reflects the foresight and advantages of TCM in the means of treating asthma. Tinglizi shows a very excellent anti-inflammatory properties and is effective against improving lung damage (Zhao et al., 2021). Dazao are responsible for protecting the spleen and stomach and enhancing protective intestinal functions (Zhao et al., 2021). Tingli Dazao Xiefei Decoction (TD), which

is composed of the combination of the two, is a classical and famous formula for the treatment of asthma in TCM (Su and Xiong, 2021), and its clinical efficacy is clear, but there are few systematic evaluations of the efficacy of asthma, and the mode of action of its improving effect on the lung and intestine is unclear.

We chose the SD rat model of asthma, which well mimics the pathological features of human asthma, as the study subject (Karol, 1994; Aun et al., 2017; Han et al., 2018). In the rat asthma model, we first synchronized our observations on the basal function and apparent inflammation in the lung and intestine. In terms of lung-related indicators, we observed a significant increase in the airway hyper-responsiveness index, the number of coughs and wheezes in asthmatic rats, along with a large number of inflammatory cell infiltrates and alveolar wall damage around the lung bronchioles. This is generally consistent with clinical asthma patients (Glaab et al., 2007; Glaab and Braun, 2021), indicating that we have successfully established an asthmatic rat model. In terms of intestine-related indices, asthmatic rats showed significant impairment of basic intestinal function, as well as shortened intestinal villi in colonic tissues with a large infiltration of inflammatory cells. This demonstrates a high correlation between the lung and the intestine and clarifies that the effects of asthma on lung damage can spill over to affect the intestine. To study the effects of TD on asthmatic lung and intestine, we selected the optimal dose of TD obtained from the pre-experimental screening of our group. The results showed that these symptoms were significantly improved by our administration of TD. This demonstrates systematically that the benefits of TD on asthma are simultaneous in both the lung and the intestine, fully illustrating the extraordinary efficacy of TD on asthma.

NO and CO are two gas signaling molecules that are closely related to the physiological and pathological functions of the lung and intestine (Bjørnkleit and Jenssen, 1980). A review of the literature reveals that the changes of NO and CO in asthma, but there are few studies discussing NO and CO together (Bukhari et al., 2015; Zhao et al., 2016). The lung and intestine exchanged gases all the time, and as NO and CO are two active gas signaling molecules, we believe that there must be mutual crosstalk between them. Further reviewing the literature, there is a NO-CO metabolic cycle between NO and CO, and CO can feedback inhibit the increase of NO in the pathological state (Chung et al., 2008).

This has led us to think about the pathogenesis of asthma in a new way. Does the NO–CO metabolic cycle play an important role in the progression of asthma? Can the treatment of asthma start from improving the lung and intestinal NO–CO metabolic cycle? Based on this, we conducted an in-depth study and exploration of the mechanism of TD to improve asthma around the possible disturbance of lung and intestinal NO–CO metabolism in asthma. The results are consistent with our conjecture that disorders of CO–NO metabolism occur to both the lung and intestine of asthmatic rats, and that NO–CO metabolic disorders in the lung and intestine are ameliorated by the administration of TD. This suggests that asthma is associated with disorders of lung and intestinal NO–CO metabolism, and TD can target the lung and intestine to provide an ameliorative effect. In addition, it is noteworthy that our positive drug, dexamethasone, seems to improve the lung and intestinal effects only on the lung, and it does not seem to have an effect on the intestine. The side effects of long-term dexamethasone administration had been previously reported in the literature (Zeng et al., 2016), and we speculate that this may be related to the mechanism of action by which it exerts its effects. In our experiments, we observed that dexamethasone did not significantly improve the levels of CO and HO-1, which are critical of NO–CO metabolism. This suggests that the side effects of dexamethasone may be closely related to NO–CO metabolism, which also indicates from the sidelines that the balance of NO and CO is pivotal to the treatment of asthma. The restoration of lung and intestinal NO–CO metabolic disorders in asthmatic rats by TD fully demonstrates the advantages of TCM for asthma treatment.

After we clarified that TD has the effect of ameliorating the disorder of NO–CO metabolism in asthma, we explored the mechanism surrounding NO–CO metabolism with a view to further elucidating the amelioration mechanism of the possible chain reaction caused by TD on NO–CO metabolic disorders. A review of the literature revealed that CD4⁺ T cells and CD8⁺ T cells are differentially regulated by NO, suggesting a correlation between NO and immune cells (Hoffman et al., 2002). Could immune cells in the lung and intestine of asthmatic rats be affected by NO–CO metabolism disorders? We examined the changes of immune cells in asthmatic rats and showed that CD4⁺T cells/CD8⁺T cells showed a significant imbalance in the ratio in lung and intestine simultaneously, and the changes of CD4⁺T cells cell population were more obvious. CD4⁺ T cells are an important class of immune regulatory cells in the immune system (Shi et al., 2011), and the ratio of their differentiated Th1, Th2, Th17, and Treg cells is constantly changing after being affected by disease, which in turn causes abnormal alterations in the level of inflammation (Chen et al., 2017). This was verified by our results, where a significant dysregulation of the ratio of Th1 cells/Th2 cells and Th17 cells/Treg cells was observed in the lung and intestine of asthmatic rats, along with severe inflammatory responses observed in pathological sections and WB findings. The dysregulation of lung and intestinal immune cells was corrected after TD administration. This implied that disorders of lung and intestinal NO–CO metabolism in asthma can provoke immune cell disorders, and TD can target lung and intestinal immunity for simultaneous amelioration. In addition, NO is closely related to intestinal homeostasis and the development of intestinal diseases (Savidge, 2014), and some intestinal microbiota can cause changes in the activity of nitric oxide synthase, which is closely related to NO production, by affecting related hormones or pathways (Savidge, 2014). This means that changes in intestinal flora are closely related to NO metabolism. Our previous experiments have identified the regulatory role of TD on NO–CO metabolism, so does TD have an effect on the intestinal flora of asthma? Based on this, we examined the changes of asthmatic with intestinal flora. In-depth analysis of the results of the intestinal flora, we clarified the restoration of intestinal flora diversity of asthmatic rats by TD based on alpha diversity analysis, beta diversity analysis, and LDA-LEfSe analysis, and found five differentially altered probiotic species at the genus level: *g.Lachnospiraceae_NK4A136*, *g.Pre-votellaceae_Ga6A1*, *g. UCG_005*, *g. Intestinimonas* and *g. Colidextribacter*. A review of the literature revealed that *g.Lachnospiraceae_NK4A136* has

anti-inflammatory and intestinal mucosal repair effects and can effectively alleviate intestinal inflammation (Zhong et al., 2020), *g.Pre-votellaceae_Ga6A1* has a role in maintaining host immune homeostasis and intestinal mucosal repair (Guo et al., 2016; Zhu et al., 2020), *g. UCG_005* is associated with digestive function (Vojinovic et al., 2019), *g. Intestinimonas* and *g. Colidextribacter* are associated with immune dysregulation (Mager et al., 2020; Li et al., 2022). All the results of this part point in a clear conclusion that the TD has a significant modification effect on the intestinal flora of asthmatic rats. In order to clarify the material basis of the herbal compound, we also identified the components of TD based on HPLC-MS, which resulted in the identification of 39 compounds containing flavonoid glycosides, glucans, organic acids, etc., which may be the material basis for TD to exert its ameliorating effects on asthmatic lung and intestinal injury.

Summarizing the results of our animal experiments, we clearly showed that TD can target lung and intestinal NO–CO metabolism to improve lung and intestinal injury in asthmatic rats. Then what is the mode of action of TD on lung and intestinal injury? Does it act directly on intestinal damage to improve the effect? In animal, the organism is a whole, and its systems are relatively independent of each other and interact with each other, which makes it difficult to explain the mode of action of the drug. To solve this problem, we have to separate the lung and the intestine, which requires cell co-culture techniques. Cell co-culture refers to the cultivation of 2 or more cells (from the same or different tissues) in the same culture system. The cell co-culture technique has the feature that it can mimic the *in vivo* environment to a great extent, and it can observe the cell-cell and cell-culture environment interactions well. Based on this, we selected lung cells BEAS-2B and intestinal cells NCM-460 to establish a model of intestinal injury caused by lung injury using co-culture chambers. BEAS-2B are human lung bronchial epithelial cells, which are commonly used to study respiratory-related diseases (Yamamoto et al., 2016; Huang et al., 2021). NCM-460 cells are human normal colonic mucosal epithelial cells, which constitute the intestinal mucosal epithelium that is key to the intestinal barrier and are commonly used to study various biological functions of the intestine (Gong et al., 2022; Zhou et al., 2022). Based on this, we established and evaluated a model of intestinal injury induced by lung injury, and showed that BEAS-2B, the lung cells stimulated by LPS, were able to transmit the injury to NCM-460, the intestinal cells. This showed that we successfully simulated the process of lung injury causing intestinal injury in the organism, which indicated that we successfully established an *in vitro* cell model of lung injury causing intestinal injury, which is the BEAS-2B /NCM-460 co-culture system.

Based on the BEAS-2B /NCM-460 co-culture system, we can then discuss on the mode of action and in-depth mechanism of TD on lung and intestinal injury. Firstly, in order to verify the changes of NO–CO metabolism in animal experiments, we examined the changes of inflammation and NO–CO metabolism in BEAS-2B /NCM-460 co-culture system, and the results showed that lung cells BEAS-2B showed a significant inflammatory response and disturbance of NO–CO metabolism after LPS stimulation. At the same time, simultaneous changes in intestinal cells NCM-460 occurred due to the effect of lung cells BEAS-2B injury. This allowed us to determine that the same NO–CO metabolic disturbances as in asthmatic rats were also seen in our BEAS-2B /NCM-460 co-culture system. Then, in order to deeply investigate the effects of NO–CO metabolic disorders on intestinal cells NCM-460, further review of the literature revealed that NO correlates with intestinal barrier damage (Chen et al., 2003) and CO correlates with intestinal barrier protection (Z. H. Liu et al., 2010). Colonic mucosal epithelial cells NCM-460 are the key to constitute the intestinal barrier, and the impairments to the intestinal barrier are usually accompanied by increased cell permeability, which in turn leads to severe oxidative stress, which forms a vicious circle leading to more severe damage to the intestinal barrier. So, can intestinal cells NCM-460 be affected by lung cells BEAS-2B damage, which in turn leads to intestinal barrier damage? If so, is the mechanism of intestinal barrier damage through cell permeability

and oxidative stress pathways? With our questions, we examined the changes of intestinal cells NCM-460, and the results showed that the BEAS-2B damage of lung cells caused by LPS caused the damage of cell permeability and oxidative stress in intestinal cells NCM-460, which means that there was a serious intestinal barrier damage. After clarifying the effects on intestinal cells NCM-460, we investigated the role and mode of TD intervention. The experimental results revealed that in the BEAS-2B /NCM-460 co-culture system, TD could only produce ameliorative effects on intestinal cells NCM-460 inflammation, NO–CO metabolic disorders, cell permeability and oxidative stress in the presence of lung cells BEAS-2B. This suggests that the ameliorative effect of TD on intestinal cells require the participation of lung cells and are exerted on intestinal cells injury by modulating lung cells injury.

To summarize our experimental studies. We first demonstrated that disordered NO–CO metabolism is an important cause of lung and intestinal injury in asthma, and then we systematically evaluated and thoroughly investigated the mechanism of TD's ameliorating effect on asthmatic lung and intestinal NO–CO metabolic disorders *in vivo*. The experiments confirmed that differentiated from our positive drug dexamethasone, TD can simultaneously alleviate asthmatic lung and intestinal NO–CO metabolic disorders, and thus ameliorate asthmatic lung and intestinal inflammatory damage, immune homeostasis, and intestinal flora. NO–CO metabolism is an important mechanism of TD for asthma. *In vitro* experiments, we established the first model of lung injury causing intestinal injury, and based on this model, we investigated the mode of action and mechanism of the ameliorative effect of TD on lung and intestinal NO–CO metabolic disorders, and clarified that TD can indirectly alleviate intestinal inflammation, cell permeability and oxidative stress by modifying lung injury. In summary, we found that targeting asthma lung and intestinal NO–CO metabolism may be a new idea and target for asthma treatment, and TD can target asthma lung and intestinal NO–CO metabolic disorders to improve asthma in a comprehensive and multidimensional manner. This study provides new clinical ideas and insights for asthma and expands the theoretical basis for the clinical use of TD.

5. Conclusions

TD can ameliorate NO–CO metabolism in the lung and thus achieve the indirectly amelioration of NO–CO metabolism in the intestine, ultimately achieving co-regulation of lung and intestinal inflammation, immune imbalance, cellular barrier damage, oxidative stress and intestinal bacterial disorders in asthma *in vivo* and *in vitro*. Targeting lung and intestinal NO–CO metabolic disorders in asthma may be a new therapeutic idea and strategy for asthma.

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CRediT authorship contribution statement

Yuan Ruan: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Visualization, Investigation. **Pei-Pei Yuan:** Conceptualization, Methodology, Software, Data curation, Writing – original draft. **Pan-Ying Li:** Visualization, Investigation. **Yi Chen:** Visualization, Investigation. **Yang Fu:** Visualization, Investigation. **Li-**

Yuan Gao: Supervision. **Ya-Xin Wei:** Supervision. **Ya-Juan Zheng:** Software, Validation. **Sai-Fei Li:** Software, Validation. **Wei-Sheng Feng:** Writing – review & editing. **Xiao-Ke Zheng:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.116503>.

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